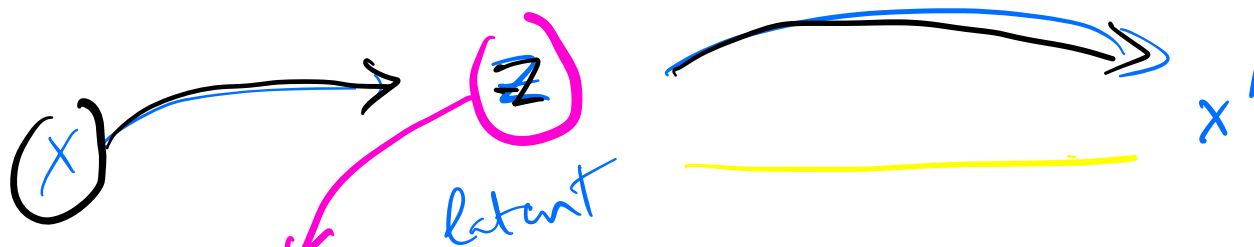


CS 6180 10/29

Nov 17 Quiz

Encoder

Decoder



$q_\phi(z|x)$
close
 $p(z)$

we can sample from $p(z)$ since $p(z)$ will be very close to $q_\phi(z|x)$ once the model is trained.

$KL(q_\phi(z|x), p(z))$

minimized ≈ 0
when model has converged

Generative Adversarial Networks
(GANs)

really good in terms of generating fake images.

Generator

Discriminator

Xiwei

Julia

↓
paint an exact replica
of the Mona Lisa

working against each other
Xiwei's goal is to draw amazing
paintings that are identical to the
Mona Lisa.



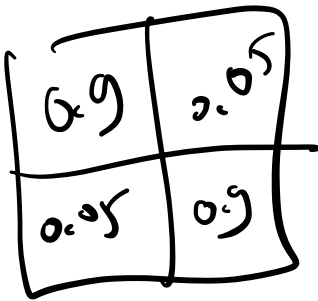
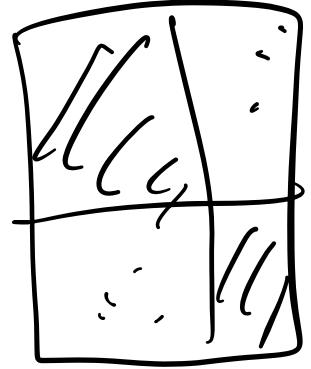
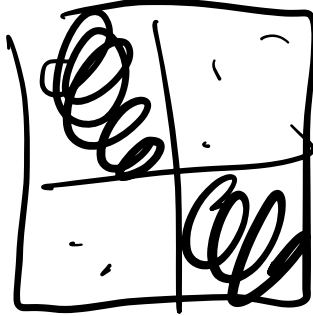
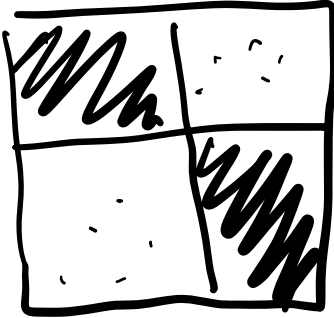
Nop

Julia: what is that?
Is that supposed to be the

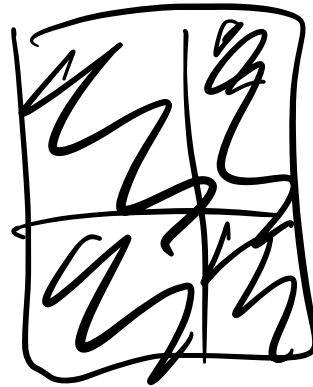
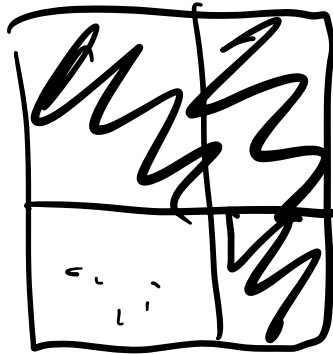
Mona Lisa?

Example

Real images



fake
images



Building the discriminator

0.8	0.1
0.1	0.8

1.4

0.9	0.05
0.05	0.9

1.7

0.1	0.9
0.1	0.6

-0.3

simple metric

a_{11}	a_{12}
a_{21}	a_{22}

$$a_{11} + a_{22}$$

$$-a_{12} - a_{21}$$

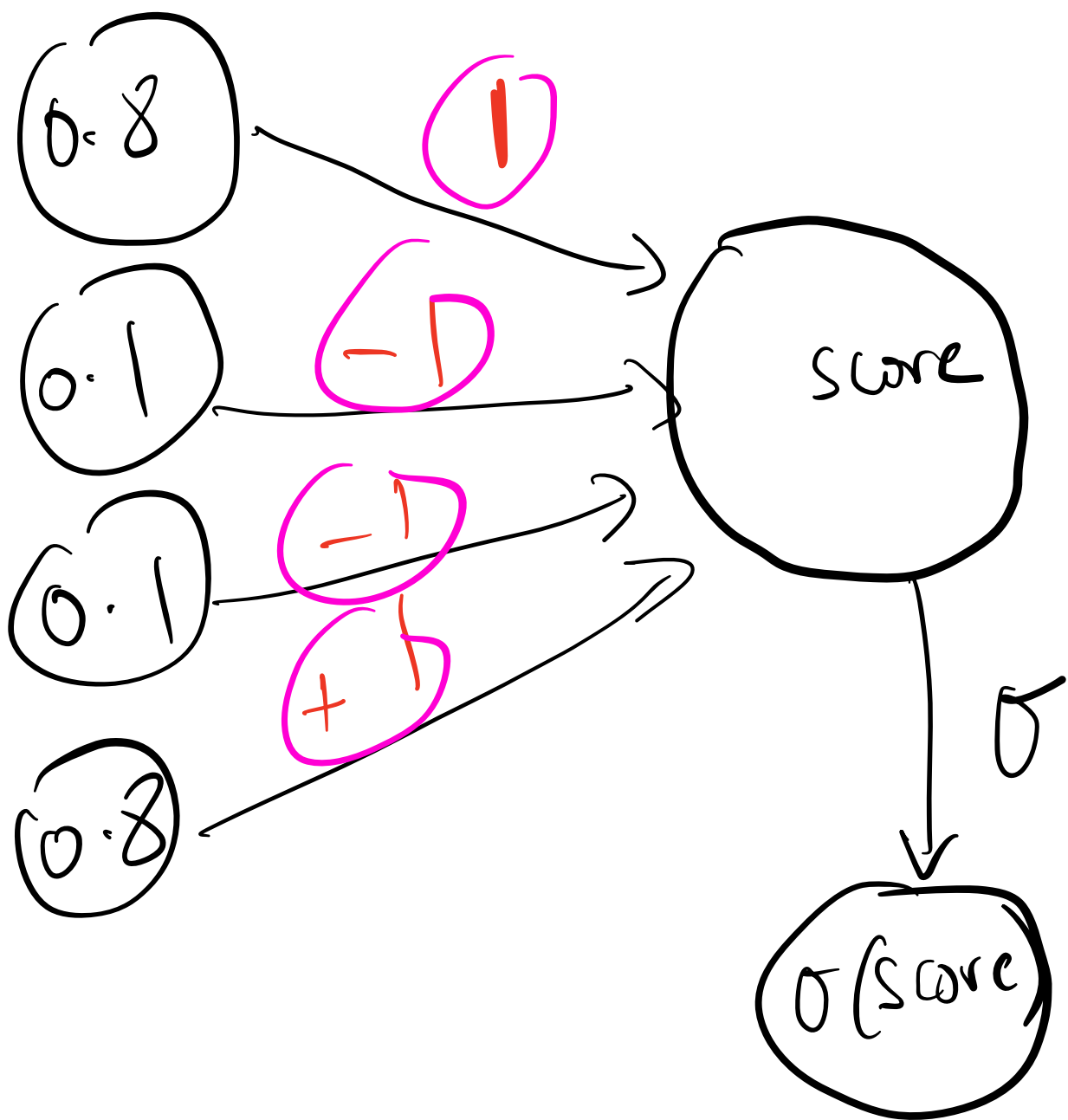
need a threshold to
indicate whether it's
a real image or not

threshold $> 1 \rightarrow$ real
else $\rightarrow$ noise

not ideal though

$\sigma(\text{score})$ $\nearrow \approx 1$ if score is large
 $\searrow \approx 0$ if score is really negative

0.8	0.1
0.1	0.8



need to learn the
weights (optimal ones)

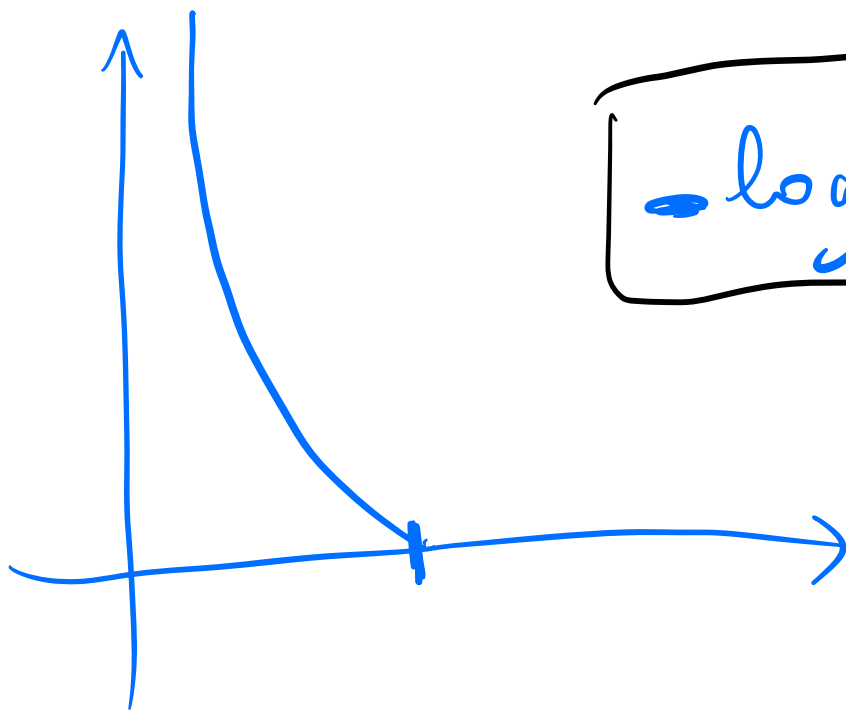
not just by eyeballing them.

Loss function to help us figure out the optimal weight

* label = 1 (real Image)

• prediction = 0.1 $\rightarrow$ large error

• prediction = 0.9 $\rightarrow$ small error



$-\log(\text{prediction})$

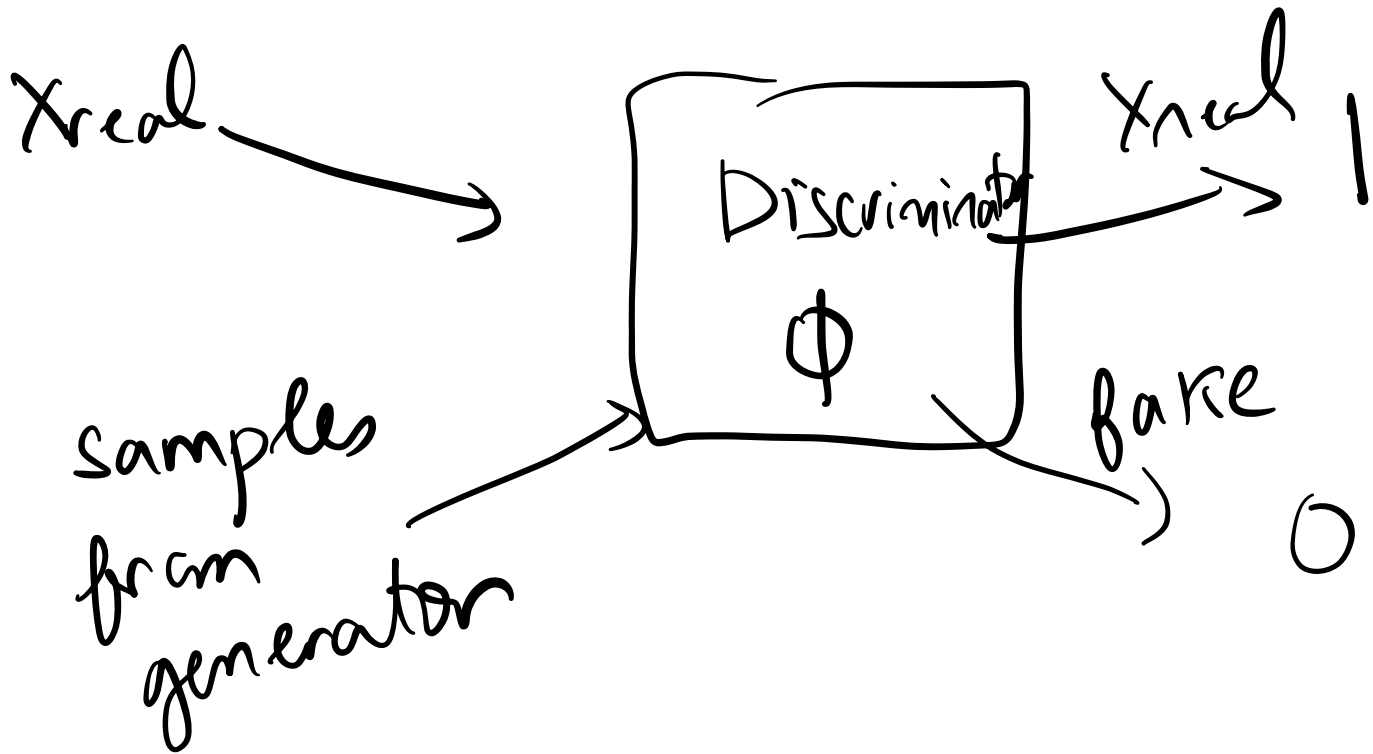
label = 0 (fake image)

• prediction = 0.1
→ small error

• prediction = 0.9
→ large error

$$-\log(1 - \text{prediction})$$

Let's build on these two error functions to figure out what the discriminator needs to do.



$$* -\log(D\phi(\underline{x_{real}}))$$

$$\text{label} = 1$$

On average, after sampling
many real images

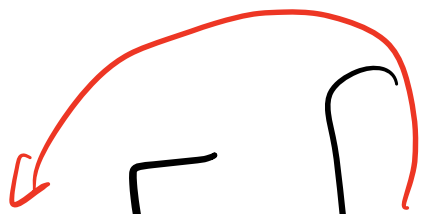
$$E_{x \sim P_{data}} [-\log(D\phi(x))]$$

$$* \text{label} = 0$$

$$\underline{G_{\theta}(z)}$$

$$-\log(1 - D_{\phi}(G_{\theta}(z)))$$

look at more than one
fake image



$$E_z \left[-\log(1 - D_{\phi}(G_{\theta}(z))) \right]$$

total loss function for discriminator

$$L_D(\phi, \theta) = -E_{x \sim p_{\text{data}}} [\log(D_{\phi}(x))]$$

$$- E_z [\log(1 - D_{\phi}(G_{\theta}(z)))]$$

What if I was the generator?

* Generator wants to fool the discriminator.

label = 0

prediction = 0.95
of
the
discriminator

} great scenario
for the
generator

$$-\log(D_{\phi}(G_{\theta}(z)))$$

$$L_G^{(1)}(\phi, \theta) = -\mathbb{E}_z [\log(D_{\phi}(G_{\theta}(z)))]$$

better

maximize $\boxed{-\log(1 - D_\phi(g_\theta(z)))}$

$D \approx 1 \rightarrow -\log(1-D)$ large

$\rightarrow D \approx 0 \rightarrow -\log(1-D)$ small

minimize $\log(1 - D_\phi(g_\theta(z)))$

$$L_G^{(2)} = \mathbb{E}_z [\log(1 - D_\phi(g_\theta(z)))]$$

struggles with
vanishing gradients

(will prove in next HW)

Exercise

$$LD(\theta, z)$$

$$= -E_{x \sim p_{data}} [\log D\phi(x)]$$

$$- E_{x \sim p_{gen}} [\log (1 - D\phi(x))]$$

Show that the optimal

discriminator:

$$D^* = \frac{p_{data}(x)}{p_{gen}(x) + p_{data}(x)}$$

$$E_{x \sim p_{data}(x)} [\log D\phi(x)]$$

$$= \sum_x P_{\text{data}}(x) \log D_{\phi}(x)$$

$$E_{x \sim P_{\text{gen}}} [\log (1 - D_{\phi}(x))]$$

$$= \sum_x P_{\text{gen}}(x) \log (1 - D_{\phi}(x))$$

$$L_D = - \sum_x P_{\text{data}}(x) \log (D_{\phi}(x)) - \sum_x P_{\text{gen}}(x) \log (1 - D_{\phi}(x))$$

$$= - \sum_x \left[P_{\text{data}}(x) \log (D_{\phi}(x)) + P_{\text{gen}}(x) \log (1 - D_{\phi}(x)) \right]$$

$$\left. \frac{\delta L_D}{\delta D} \right|_{D^*} = 0$$

$$\Rightarrow \sum_x \left[P_{\text{data}}(x) \frac{1}{D\phi(x)} + P_{\text{gen}}(x) \frac{-1}{1-D\phi(x)} \right] = 0$$

$$\Rightarrow \sum_x \left[\frac{P_{\text{data}}(x)}{D\phi(x)} - \frac{P_{\text{gen}}(x)}{1-D\phi(x)} \right] = 0$$

if $D\phi(x) = \frac{P_{\text{data}}(x)}{P_{\text{gen}}(x) + P_{\text{data}}(x)} \quad \forall x$

equivalence to saying:
require every term in the sum to
be 0.

$$\frac{P_{\text{data}}(x)}{D\phi(x)} - \frac{P_{\text{gen}}(x)}{1-D\phi(x)} = 0 \quad \underline{\underline{\forall x}}$$

$$\Rightarrow \frac{P_{data}(x)}{D_{\phi}(x)} = \frac{P_{gen}(x)}{1 - D_{\phi}(x)}$$

$$\Rightarrow \frac{1 - D_{\phi}(x)}{D_{\phi}(x)} = \frac{P_{gen}(x)}{P_{data}(x)}$$

$$\Rightarrow \frac{1}{D_{\phi}(x)} - 1 = \frac{P_{gen}(x)}{P_{data}(x)}$$

$$\Rightarrow \frac{1}{D_{\phi}(x)} = \frac{P_{gen}(x) + P_{data}(x)}{P_{data}(x)}$$

$$\Rightarrow D_{\phi}(x) = \frac{P_{data}(x)}{P_{gen}(x) + P_{data}(x)}$$

$\forall x$

$\Rightarrow$ the discriminator

$$D_{\phi}^*(x) = \frac{P_{data}(x)}{P_{gen}(x) + P_{data}(x)}$$

satisfies that $\left. \frac{\partial L_D}{\partial D} \right|_{D^*} = 0$

* Is there another discriminator that satisfies

$$\sum_x \frac{p_{\text{data}}(x)}{D_\phi(x)} - \frac{p_{\text{gen}}(x)}{1 - D_\phi(x)} = 0?$$

* How do I know it is the optimal one?